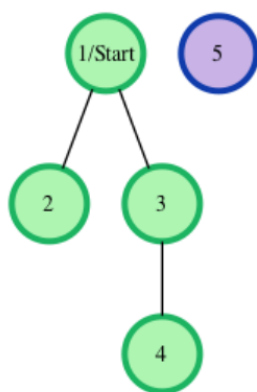


Breadth First Search: Shortest Reach

Consider an undirected graph where each edge weighs 6 units. Each of the nodes is labeled consecutively from 1 to n .

You will be given a number of queries. For each query, you will be given a list of edges describing an undirected graph. After you create a representation of the graph, you must determine and report the shortest distance to each of the other nodes from a given starting position using the *breadth-first search* algorithm ([BFS](#)). Return an array of distances from the start node in node number order. If a node is unreachable, return `-1` for that node.

The following graph is based on the listed inputs:



$n = 5$ // number of nodes

$m = 3$ // number of edges

$edges = [1, 2], [1, 3], [3, 4]$

$s = 1$ // starting node

All distances are from the start node 1. Outputs are calculated for distances to nodes 2 through 5: $[6, 6, 12, -1]$. Each edge is 6 units, and the unreachable node 5 has the required return distance of -1 .

Input Format

The first line contains an integer q , the number of queries. Each of the following q sets of lines has the following format:

- The first line contains two space-separated integers n and m , the number of nodes and edges in the graph.
- Each line i of the m subsequent lines contains two space-separated integers, u and v , that describe an edge between nodes u and v .
- The last line contains a single integer, s , the node number to start from.

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Sample Input

```
2
4 2
1 2
1 3
1
3 1
2 3
2
```

Sample Output

```
6 6 -1
-1 6
```